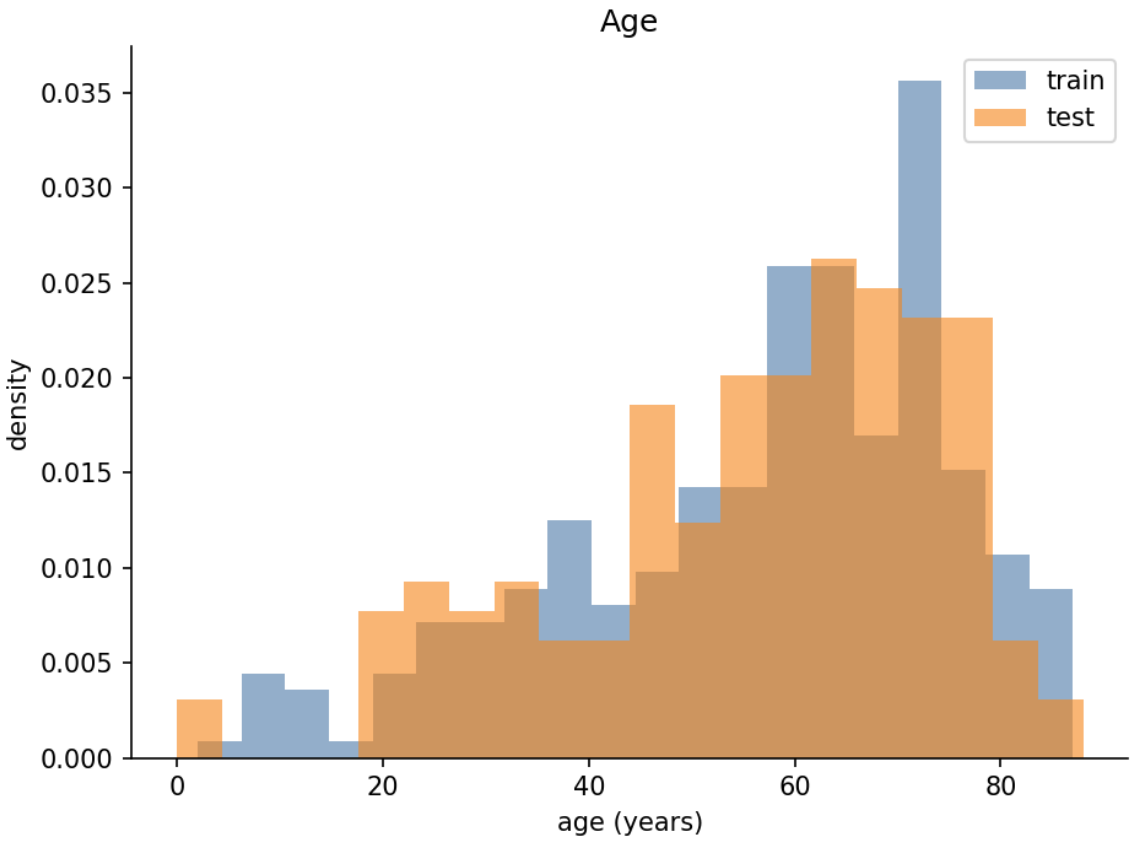
Age at specimen



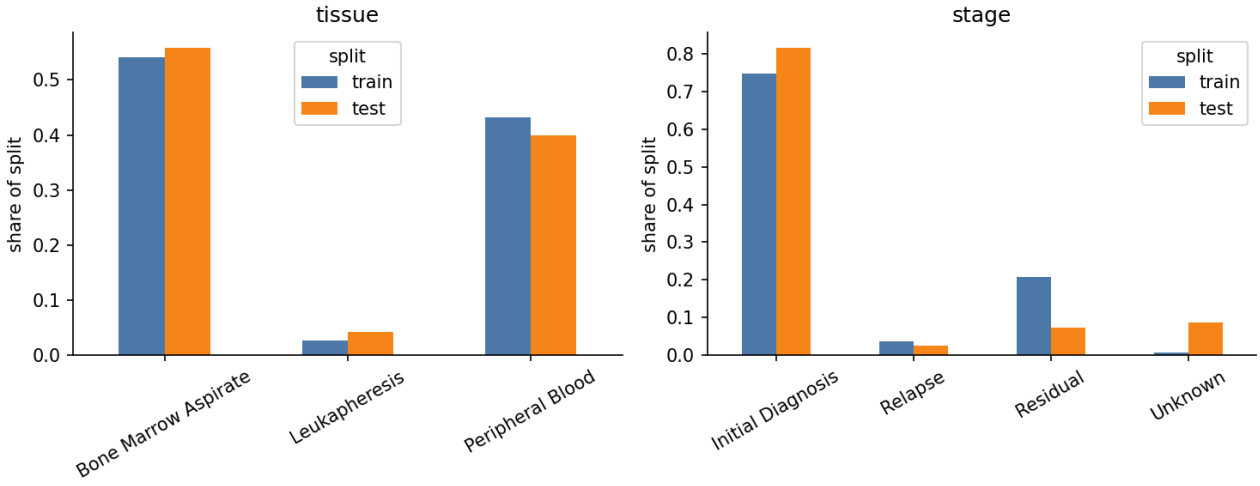
What it shows

Age distribution of train patients (blue, waves 1+2) and test patients (orange, waves 3+4). Both groups range from infants to about 88 years and peak between 60 and 75. The two shapes are close; train has a few more patients under 20.

Why it matters

Age is one of the strongest drivers of AML outcome. Matching age profiles mean the test set is a fair check of a model trained on the train set.

Specimen tissue and disease stage



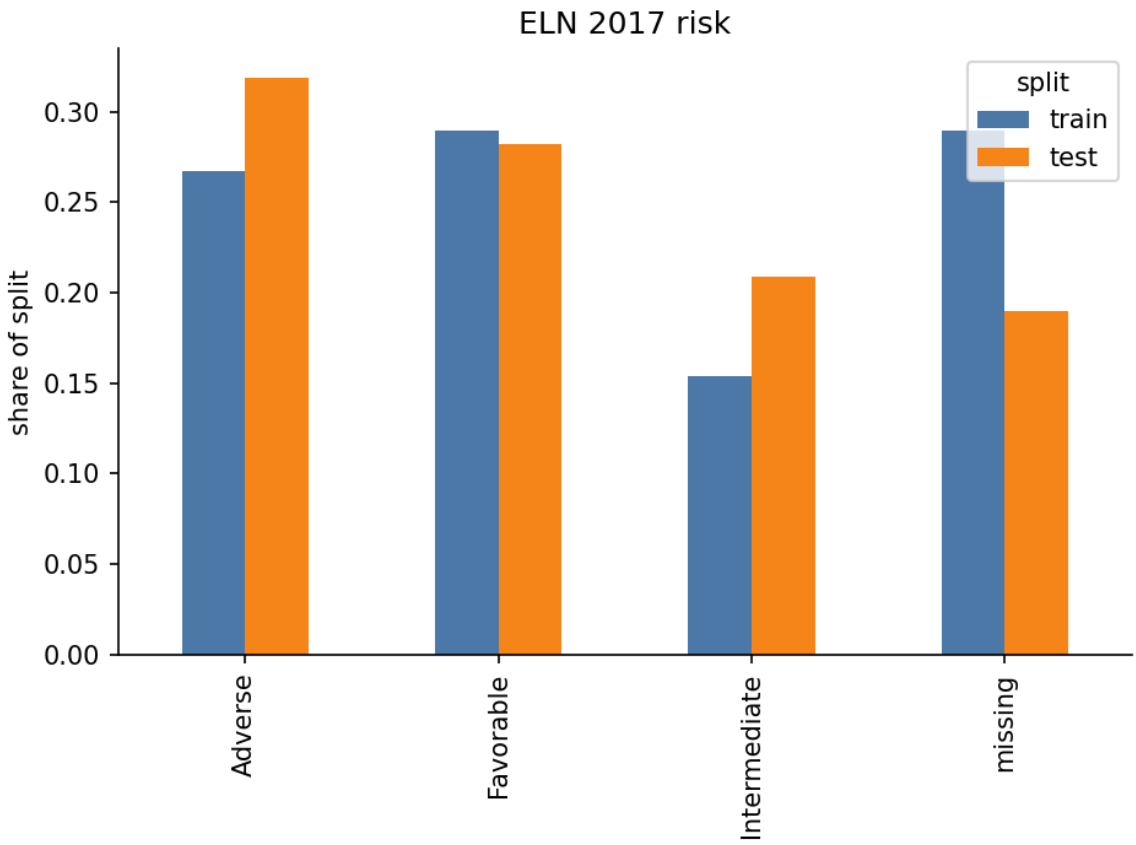
What it shows

Share of each split by where the sample came from (left) and the disease stage when it was taken (right). Bone marrow (about 55%) and peripheral blood (about 40 to 43%) are balanced across splits. Stage is not: 82% of test samples are from initial diagnosis versus 75% of train, while train has 21% residual-disease samples versus 7% in test, and about 9% of test samples have an unknown stage.

Why it matters

Tissue and stage change the mix of cell types in a sample and how the cells respond to drugs. The stage imbalance is a possible confounder, so it is worth including as a covariate or checking results within each stage.

ELN 2017 risk group



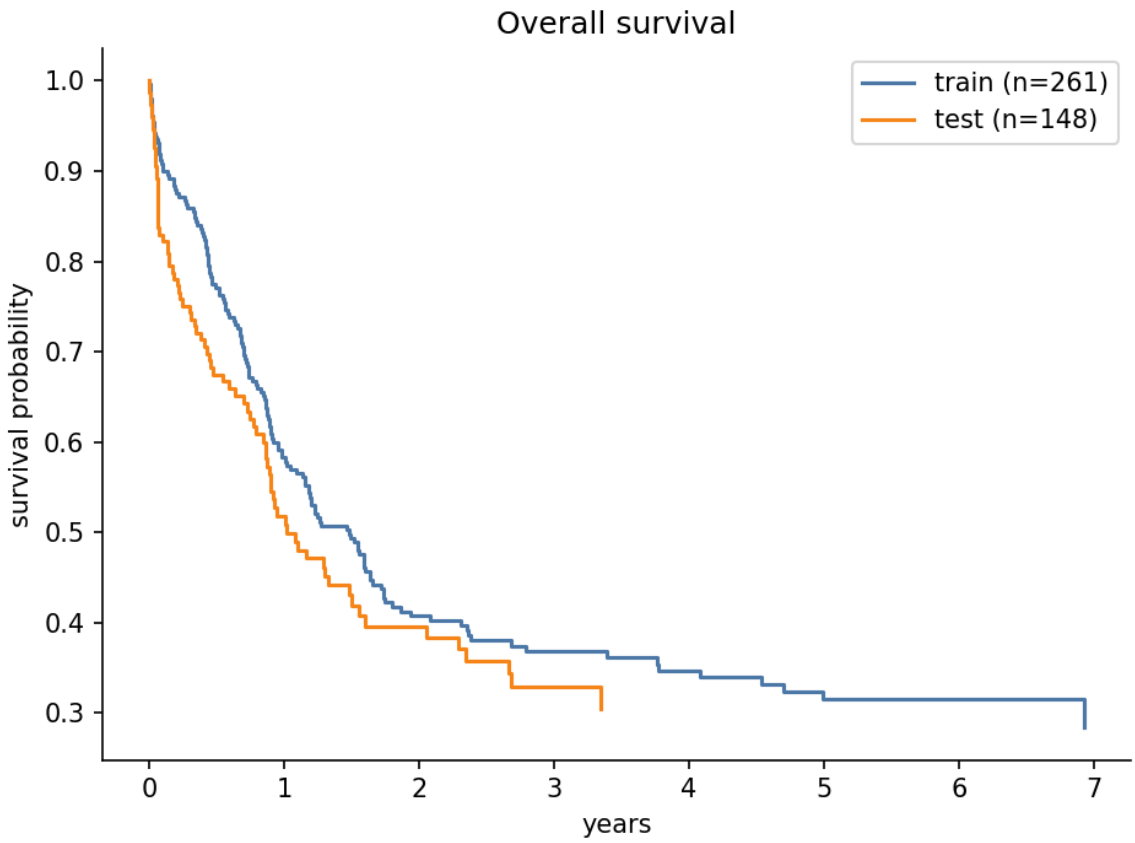
What it shows

Share of each split in each ELN 2017 risk group. Test has more adverse (32% vs 27%) and intermediate (21% vs 15%) patients, similar favorable (28% vs 29%), and fewer patients with no risk call (19% vs 29%).

Why it matters

ELN is the standard clinical risk label. It is missing for about one patient in four, so it cannot be used as a feature without a plan for the gaps, and the test set contains somewhat higher-risk disease.

Overall survival



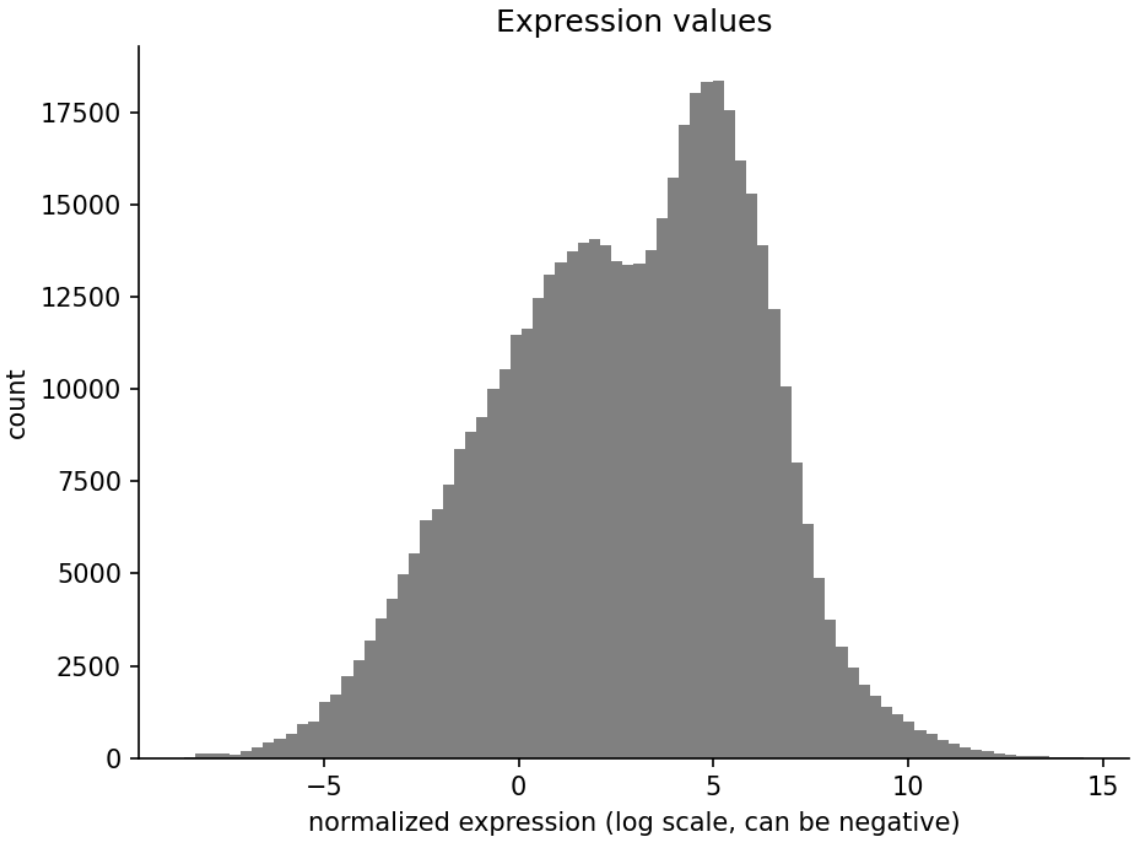
What it shows

Kaplan-Meier curves: at each death, the running 'still alive' fraction is multiplied by (1 - deaths / people still at risk). Median survival is about 1.5 years in train and 1.0 year in test. Test follow-up ends near 3.4 years; train follows patients out to about 7.

Why it matters

The test patients do worse and have shorter follow-up. For any survival analysis, compare ranking metrics (such as the concordance index) rather than raw curves, and consider limiting the time horizon to about 3 years so both groups are covered.

Distribution of expression values



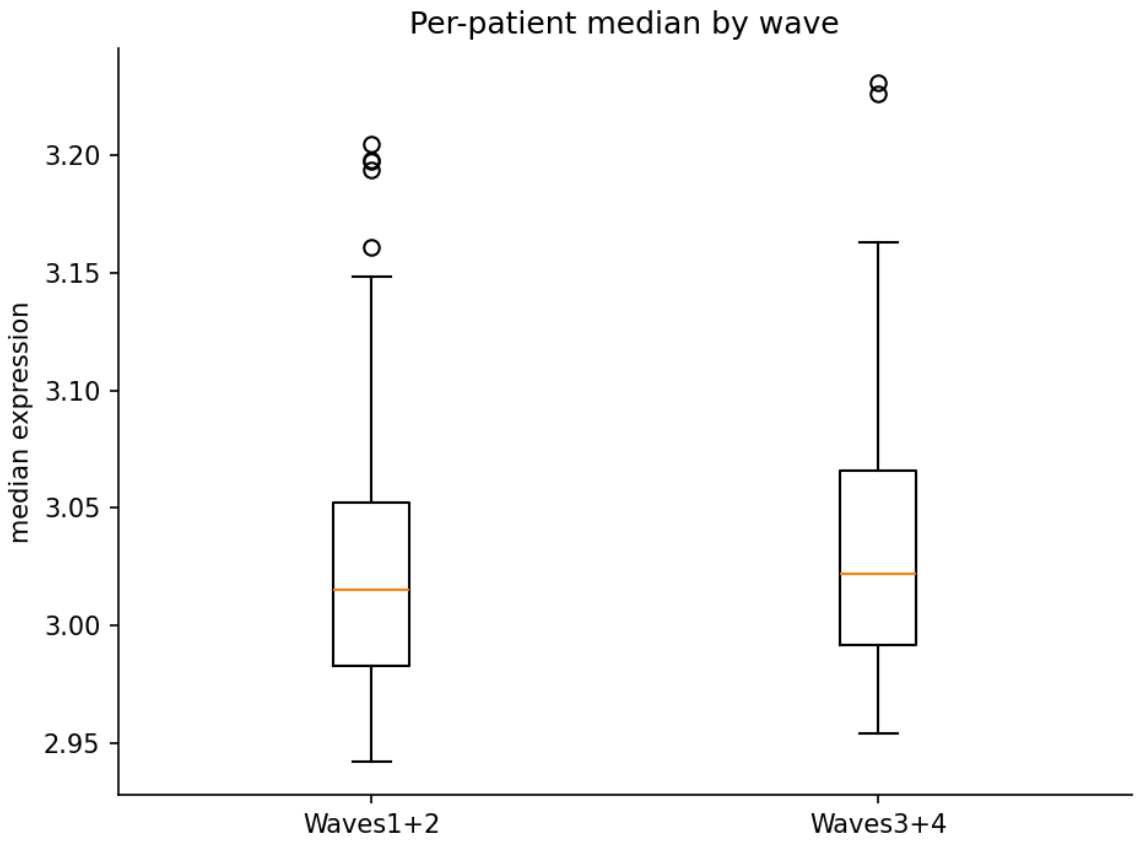
What it shows

A random sample of 500,000 values from the expression matrix. Values are log-scale normalized expression and range from about -9 to 14.5, with one hump of low or off genes near 0 to 2 and another of expressed genes near 5.

Why it matters

Negative values are normal for this normalization and are not counts. NMF needs non-negative input, so the data must be shifted or transformed first, while a VAE can use standardized values. Many genes are barely expressed, which supports keeping only variable genes.

Median expression per patient, by wave



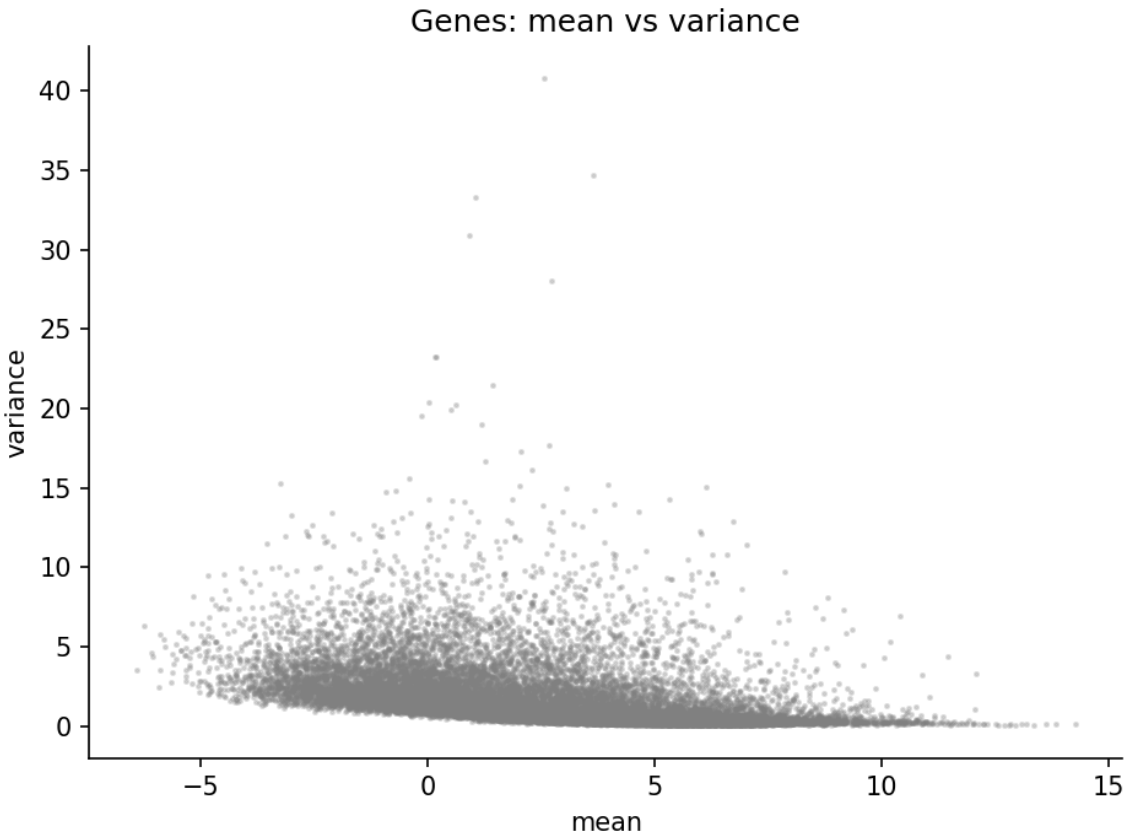
What it shows

Each box is the spread of per-patient median expression across all genes, for waves 1+2 and waves 3+4. Both are centered near 3.0 and span roughly 2.95 to 3.2.

Why it matters

There is no obvious overall offset between the waves, which is reassuring for training on one wave and testing on the other. It does not rule out gene-level batch differences, so treat it as a first check only.

Gene mean versus variance



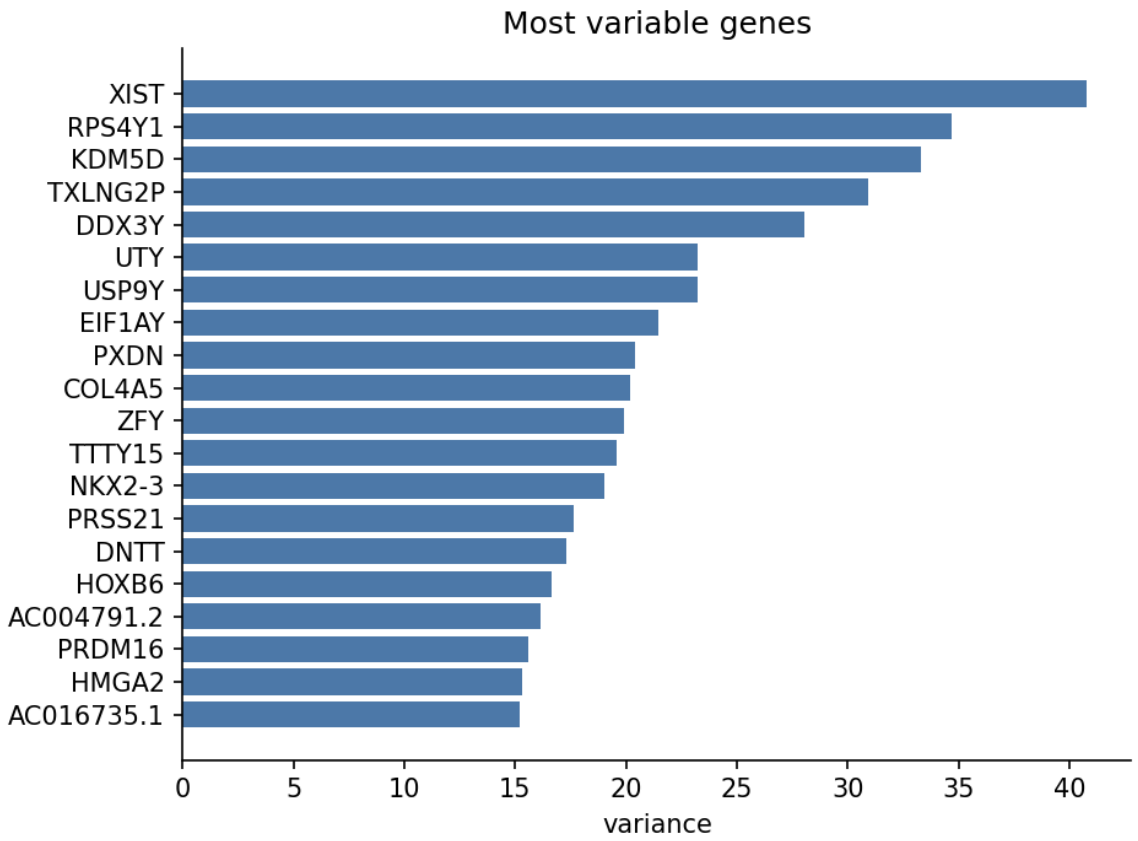
What it shows

Each dot is one of 22,843 genes. Variance is highest for genes with moderate average expression (about 0 to 4) and shrinks for highly expressed genes. A handful of genes have variance above 20.

Why it matters

Most genes barely vary across patients. Feeding a model only the top few thousand most variable genes keeps signal and cuts noise and training time.

Most variable genes



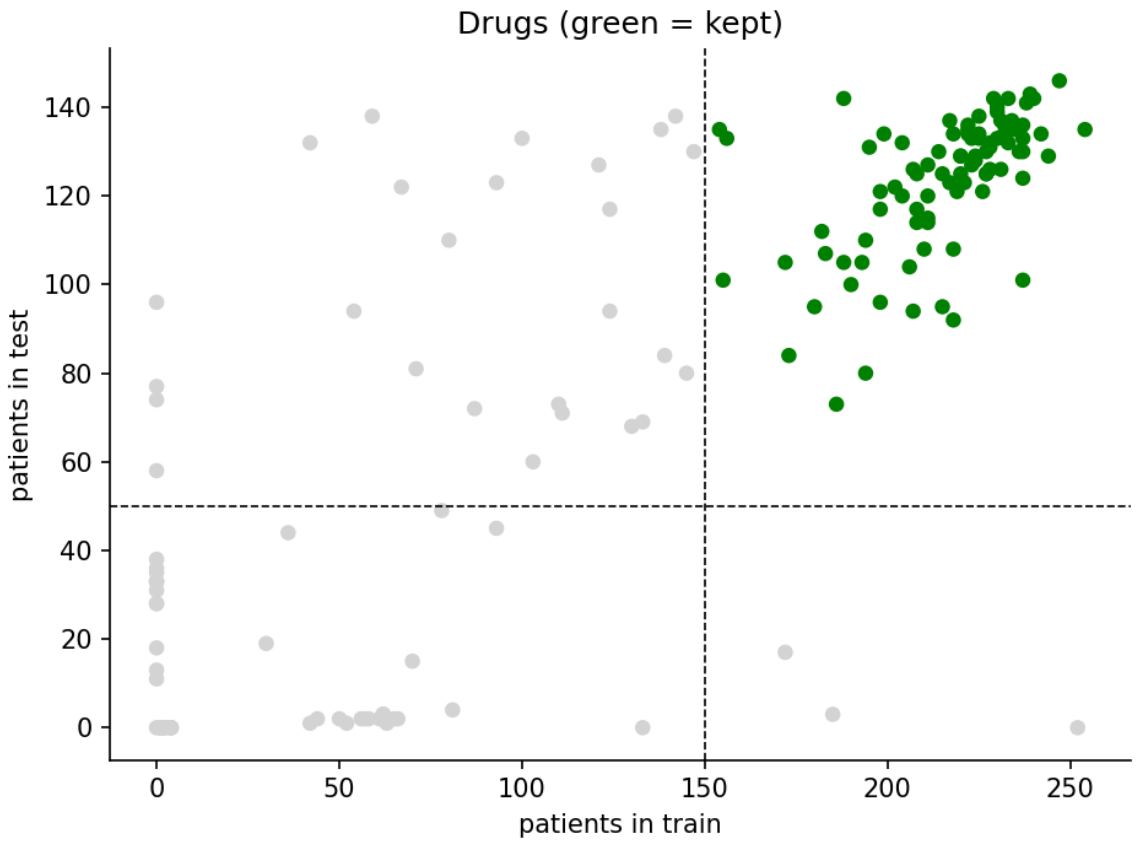
What it shows

The 20 genes with the largest variance across patients. The top of the list is dominated by sex-linked genes: XIST (active in female cells) and Y-chromosome genes such as RPS4Y1, DDX3Y, UTY, USP9Y, EIF1AY, ZFY, TTTY15 and KDM5D. Leukemia-related genes such as DNTT, HOXB6, PRDM16 and HMGA2 also appear.

Why it matters

An unsupervised model (VAE or NMF) will happily learn patient sex as a major factor, even though it has little to do with drug response. Remove sex-linked genes or regress out sex before learning latent features.

Patients per drug



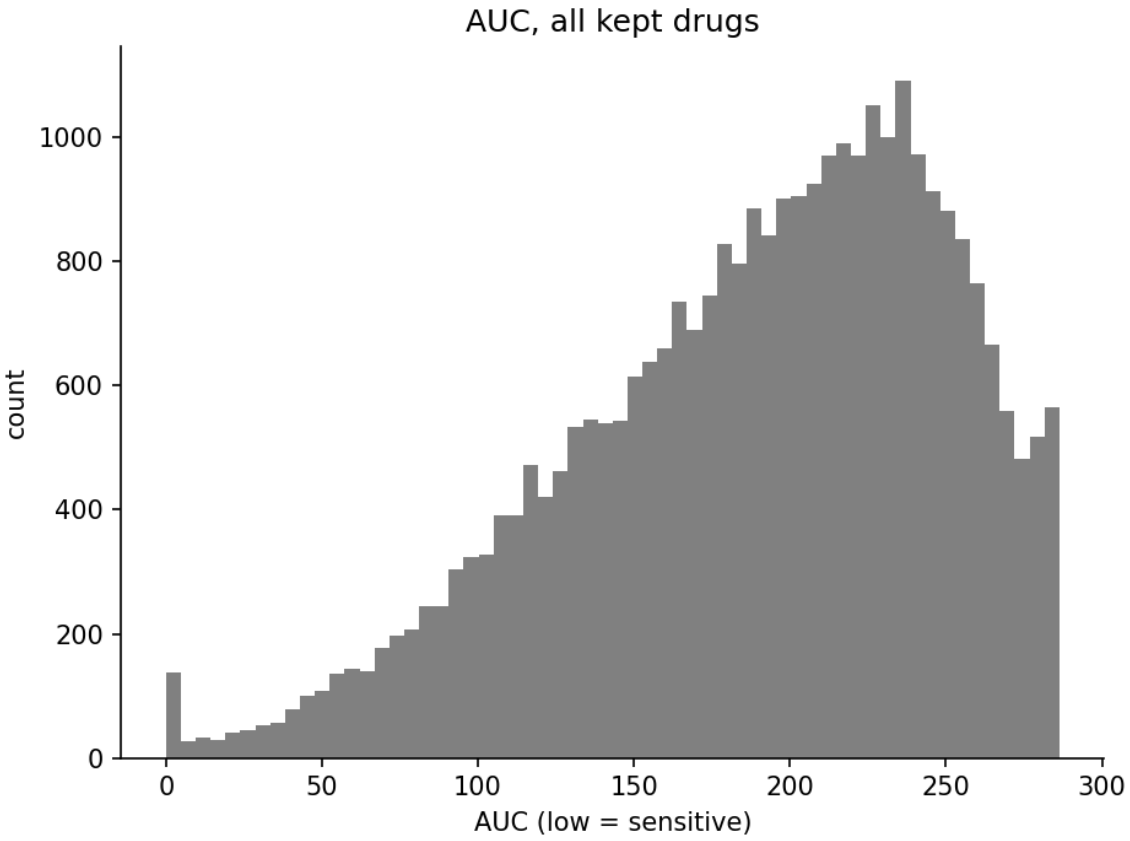
What it shows

Each dot is a drug, placed by how many train patients (x) and test patients (y) have an AUC for it. Green drugs (91) pass the thresholds of at least 150 train and 50 test patients. Many grey drugs were screened in only one wave, which is why they sit on an axis.

Why it matters

The cutoff is a choice. Venetoclax (142 train patients), Cytarabine (138), Lenvatinib (147) and a few others sit just below it, and lowering the train threshold slightly would add them.

Distribution of drug AUC



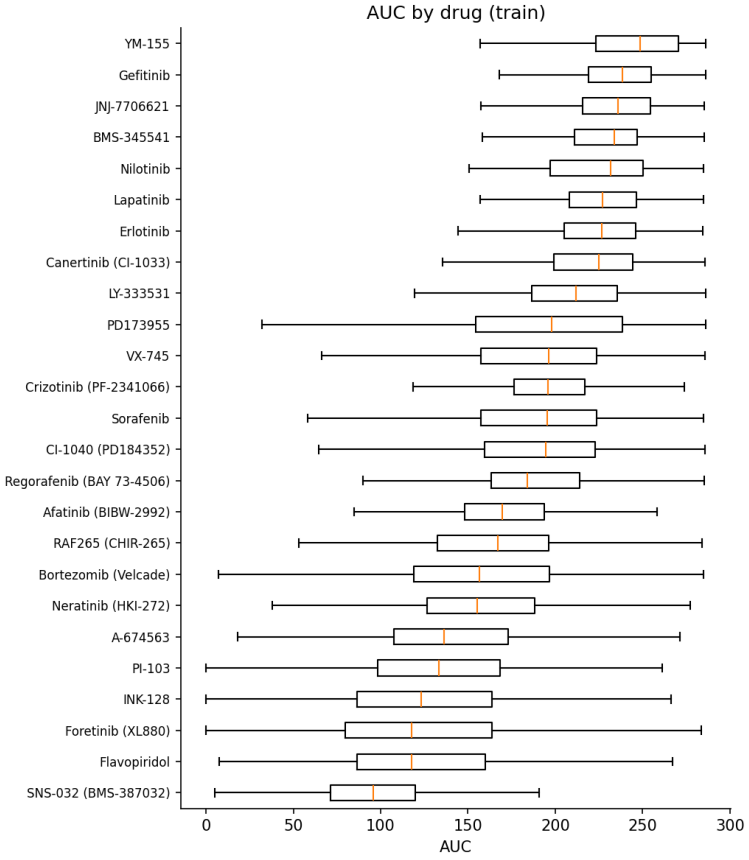
What it shows

All AUC values for the kept drugs. AUC is the area under the dose-response curve, and lower means more sensitive. Most values are high (resistant), peaking around 220 to 240, with a long tail toward sensitive, a small spike at 0 and a pile-up at the maximum near 286.

Why it matters

The outcome is skewed and capped. Many curves never fall below 50% viability, so many patients tie near the ceiling. Use rank-based metrics such as Spearman correlation alongside error metrics, and consider a transform of the target.

AUC by drug (train)



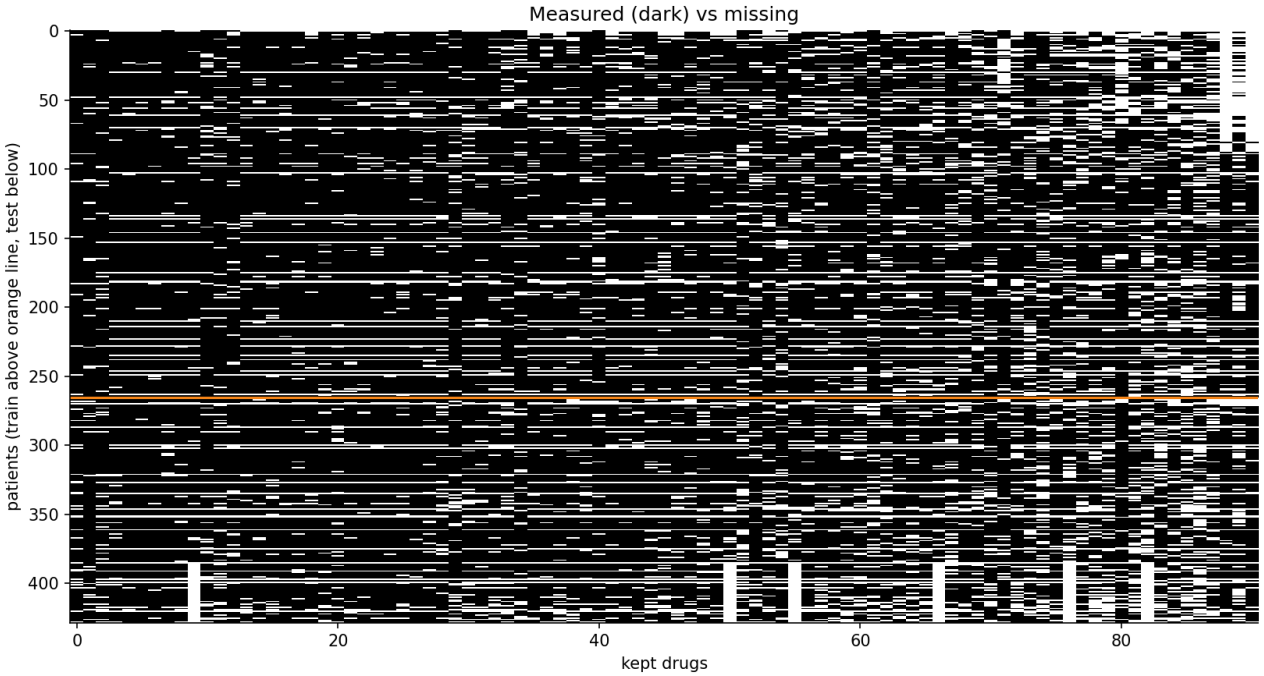
What it shows

Spread of AUC for the 25 most-measured kept drugs, sorted by median (boxes are the middle half of patients; whiskers exclude extremes). Medians run from about 95 for SNS-032 to about 248 for YM-155.

Why it matters

Drugs differ a lot in overall potency. Drugs with wide boxes (Bortezomib, PI-103, INK-128, Foretinib, PD173955) vary a lot between patients, so there is something to predict. Drugs with narrow, high boxes (Gefitinib, Erlotinib, Lapatinib) barely work on anyone and will be hard to model.

Which drugs were measured for which patients



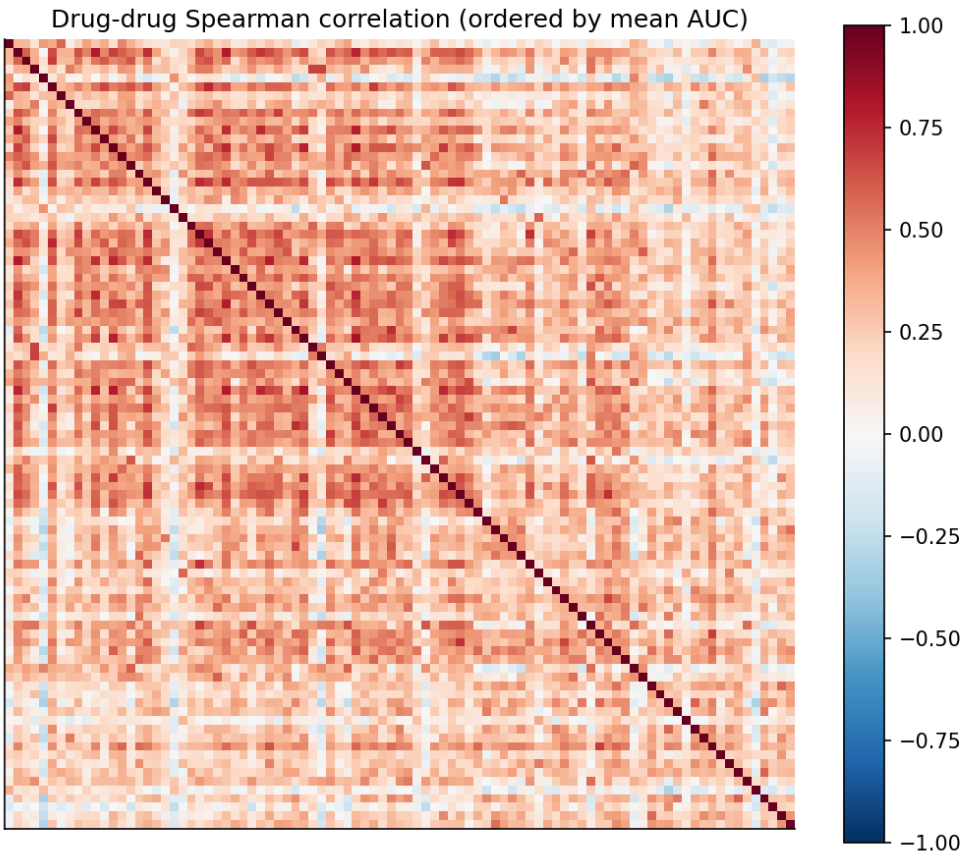
What it shows

Rows are patients (train above the orange line, test below) and columns are the 91 kept drugs. Black means measured, white means missing. Most cells are filled, but some patients are nearly blank across the row, and a few drugs have white stripes covering a block of later patients.

Why it matters

Missing data is not random. Models should handle missing drug values explicitly (for example, by masking them in the loss or fitting one model per drug), and patients with very few drugs measured may need to be dropped.

Drug-drug correlation



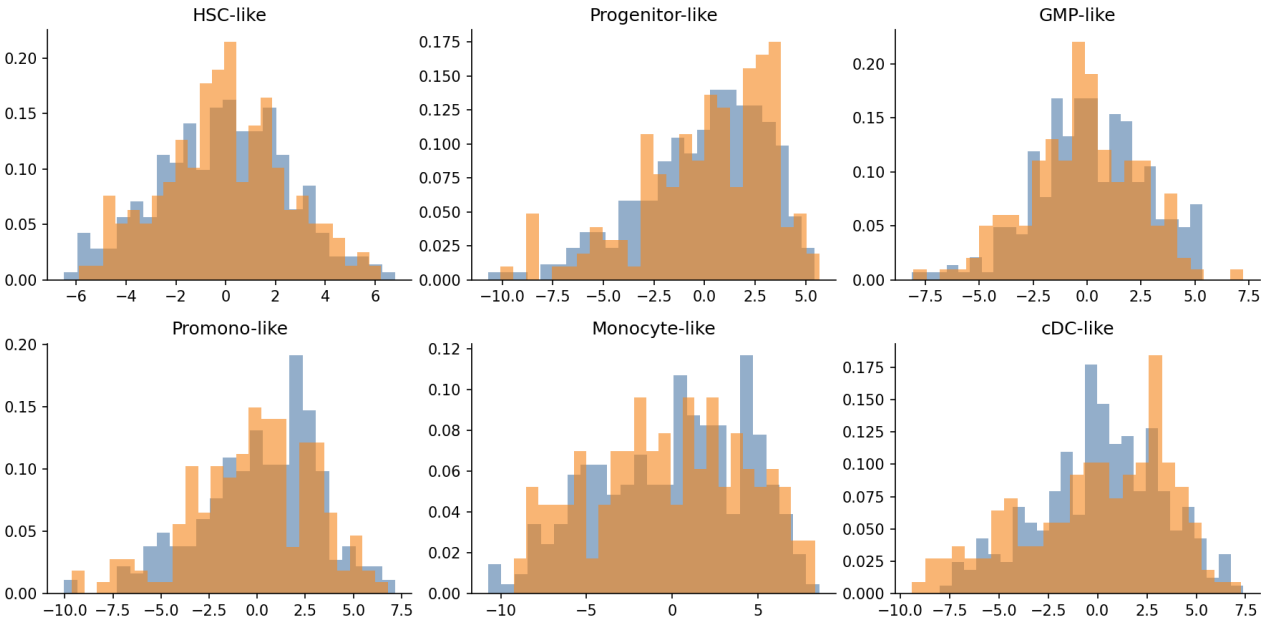
What it shows

Spearman correlation of AUC between every pair of kept drugs (train patients), with drugs ordered from most to least potent on average. The matrix is mostly red. The median pairwise correlation is 0.29, and about half of all pairs are above 0.3. A few drugs show pale columns, meaning they do not follow the shared trend.

Why it matters

A patient who is resistant to one drug tends to be resistant to many. This shared 'general sensitivity' factor can make a model look good without learning drug-specific biology, so evaluate against a baseline that captures it, or look at per-drug results after removing it.

Cell-type scores (the paper's baseline features)



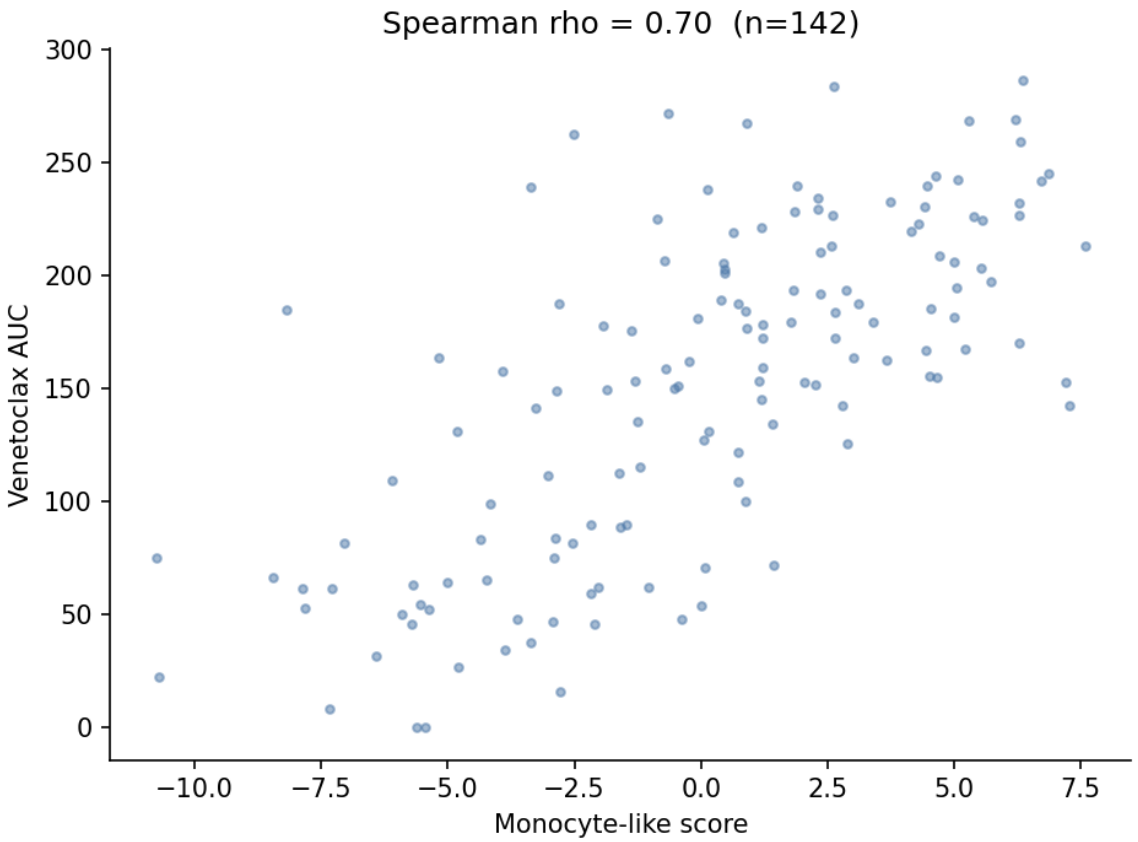
What it shows

The six van Galen cell-type scores per patient: HSC-like, Progenitor-like, GMP-like, Promono-like, Monocyte-like and cDC-like. Blue is train, orange is test. Distributions overlap well and span roughly -10 to +8.

Why it matters

These hand-built scores are the baseline that learned latent features must beat. They were fit on waves 1+2 and projected onto waves 3+4, and the similar test distributions suggest that projection worked.

Venetoclax response versus Monocyte-like score



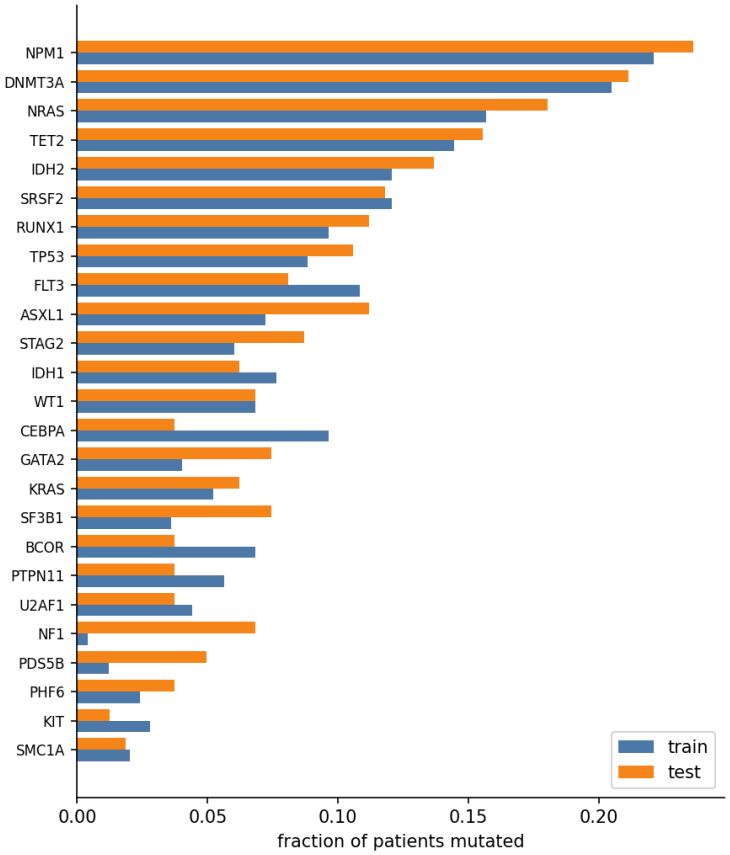
What it shows

Each dot is a train patient. A higher Monocyte-like score goes with a higher Venetoclax AUC, with Spearman correlation 0.70 across 142 patients.

Why it matters

Monocytic AML is known to resist Venetoclax, so this matches the biology. It confirms the AUC direction and the patient joins are correct, and shows that one hand-built feature already carries strong signal, which sets a high bar for new features.

Most frequently mutated genes



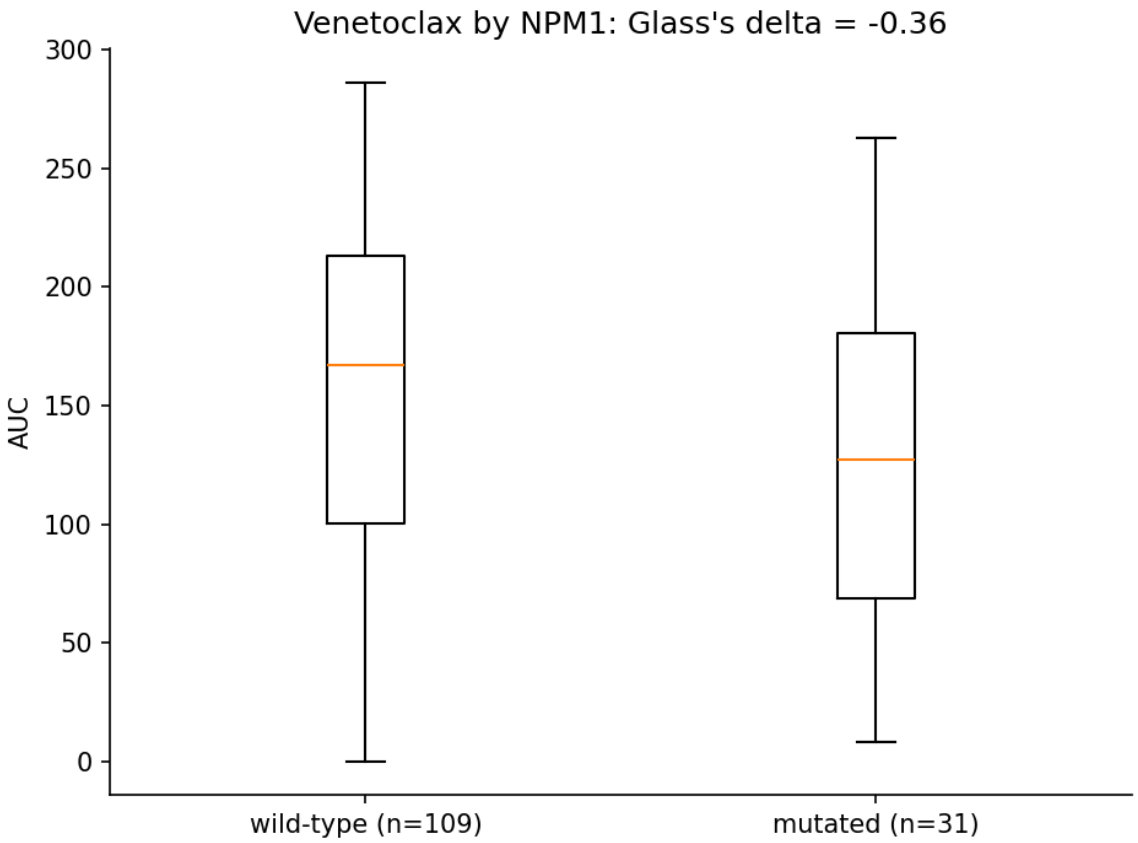
What it shows

Fraction of train (blue) and test (orange) patients with a mutation in each of the 24 most mutated genes. NPM1 (about 22 to 24%), DNMT3A (about 20 to 21%), NRAS, TET2 and IDH2 lead. Most genes match between splits, but some differ: CEBPA is about 10% in train versus 4% in test, and NF1 is under 1% versus about 7%.

Why it matters

Only a handful of genes are mutated in more than 10% of patients, so most mutations are too rare to learn drug links one at a time. Grouping genes by pathway, or using latent features, can help. The test mutation mix is slightly different from train.

Venetoclax response by NPM1 status



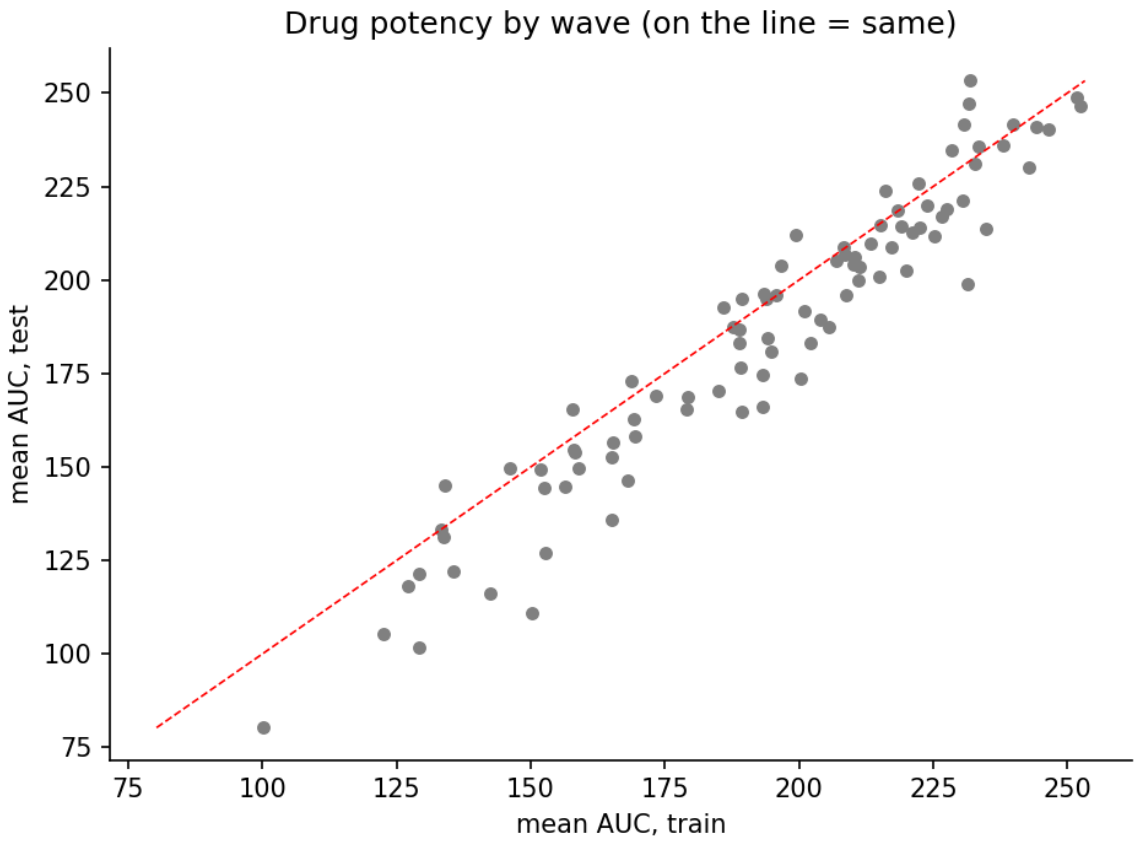
What it shows

AUC for train patients without (n=109) and with (n=31) an NPM1 mutation. Glass's delta is the difference in means divided by the wild-type spread, here -0.36. NPM1-mutated patients have a lower median AUC (about 127 versus 167), meaning more sensitive.

Why it matters

The effect is real but modest, and the two groups overlap heavily. A single mutation explains only part of drug response. This tool is for looking at one drug and one gene at a time; with many pairs, some strong-looking ones will be chance.

Drug potency in train versus test



What it shows

Each dot is a kept drug, with its mean AUC in train (x) and in test (y). Points on the red line mean the drug looks equally potent in both waves. Most drugs track the line closely, but more of them sit below it, meaning slightly lower (more sensitive) AUC in test.

Why it matters

Drug ranking is stable across waves, which is good. The small systematic shift means a model trained on waves 1+2 may over-predict AUC on waves 3+4, so judge it by rank correlation and consider recalibrating.